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* ZC416 Mathematical Foundations for
Machine Learning / Data Science

Topic: # 13 Principle Components Analysis



- In the previous lecture, we discussed dimensionality reduction using PCA
- In this lecture, we will see the use of linear algebra in practical implementation of PCA.
- We will also study the challenges encountered when PCA is used in problems of larger dimensions.
- Finally, we elaborate the key steps of PCA in practice.

REVISION OF PCA PROBLEM



- We derived the matrix $\mathbf{B}$ used in generation of lower-dimensional representation $\mathbf{z}$ and the compressed data $\tilde{\mathbf{x}}$.
- The data covariance matrix $\mathbf{S}$ was used to derive $\mathbf{B}$
- Recall that linear relationship connecting the original data $\mathbf{x}$, its low-dimensional code $\mathbf{z}$ and the compressed data $\tilde{\mathbf{x}}$: $\mathbf{z} = \mathbf{B}^T \mathbf{x}$, and $\tilde{\mathbf{x}} = \mathbf{Bz}$.

EIGENVECTOR COMPUTATION AND LOW RANK APPROXIMATION

- In the previous sections, we obtained the basis of the principal subspace as the eigenvectors that are associated with the largest eigenvalues of the data covariance matrix.

$$S = \frac{1}{N} \sum_{i=1}^N x_n x_n^T \quad (1)$$

- Equivalently we get

$$S = \frac{1}{N} X X^T \quad (2)$$

$$X = [x_1, \dots, x_N] \in \mathbb{R}^{D \times N} \quad (3)$$

- Note that X is a $D \times N$ matrix,

EIGENVECTOR COMPUTATION AND LOW RANK APPROXIMATION

- To get the eigenvalues and the corresponding eigenvectors of S , we can follow two approaches
- We can perform an eigendecomposition and compute the eigenvalues and eigenvectors of S directly.
- We can also use a singular value decomposition. Since S is symmetric and factorizes into XX^T , the eigenvalues of S are the squared singular values of X .
- Assume the SVD of X as $X = U\Sigma V^T$. Then

$$S = \frac{1}{N}XX^T = \frac{1}{N}U\Sigma\Sigma^TU^T$$

EIGENVECTOR COMPUTATION AND LOW RANK APPROXIMATION

- The columns of U are the eigenvectors of S .
- The eigenvalues λ_d of S are related to the singular values of X via

$$\lambda_d = \frac{\sigma_d^2}{N} \quad (4)$$

- This relationship between the eigenvalues of S and the singular values of X provides the connection between the maximum variance view and the singular value decomposition.



PCA USING LOW-RANK MATRIX APPROXIMATIONS

- To maximize the variance of the projected data PCA chooses the columns of U to be the eigenvectors that are associated with the M largest eigenvalues of the data covariance matrix S
- The Eckart-Young theorem offers a direct way to estimate the low-dimensional representation.
- Consider the best rank- M approximation of X defined as $\tilde{X}_M$

$$\tilde{X}_M = \operatorname{argmin}_{\operatorname{rank}(A) \leq M} \|X - A\|_2 \quad (5)$$

PCA USING LOW-RANK MATRIX APPROXIMATIONS



- The Eckart-Young theorem states that the best rank M approximation $\tilde{X}_M$ is given by truncating the SVD at the top- M singular value.

$$\tilde{X}_M = U_M \Sigma_M V_M^T$$

- U_M is an orthogonal matrix
- V_M is an orthogonal matrix
- Σ_M has M largest singular values of X as diagonal entries



- Finding eigenvalues and eigenvectors is also important in other fundamental machine learning methods that require matrix decompositions
- In theory we can solve for the eigenvalues as roots of the characteristic polynomial.
- However for matrices larger than 4 by 4 this is not possible because we would need to find the roots of polynomial of degree 5 or higher.



- However the Abel-Ruffini theorem states that there exists no algebraic solution to this problem for polynomials of degree 5 or more.
- Therefore, in practice, solve for eigenvalues or singular values using iterative methods, which are implemented in all modern packages for linear algebra
- In many applications we only require a few eigenvectors.



- It would be wasteful to compute the full decomposition, and then discard all eigenvectors with eigenvalues that are beyond the first few.
- It turns out that if we are interested in only the first few eigenvectors (with the largest eigenvalues), then iterative processes, which directly optimize these eigenvectors, are computationally more efficient than a full eigendecomposition



- In the extreme case of only needing the first eigenvector, a simple method called the power iteration is very efficient.
- Power iteration chooses a random vector x_0 that is not in the null space of S and follows the iteration for $k = 0, 1, \dots$

$$x_{k+1} = \frac{Sx_k}{\|Sx_k\|} \quad (6)$$

- This sequence of vectors converges to the eigenvector associated with the largest eigenvalue of S .



- 1 In order to do PCA, we need to compute the data covariance matrix.
- 2 In D dimensions, the data covariance matrix is a $D \times D$ matrix.
- 3 Computing the eigenvalues and eigenvectors of this matrix is computationally expensive as it scales cubically in D .
- 4 Therefore, PCA, as we discussed earlier, will be infeasible in very high dimensions.
- 5 In the following, we provide a solution to this problem for the case that we have substantially fewer data points than dimensions, i.e., $N \ll D$



- Assume we have a centered dataset $x_1, \dots, x_N, x_n \in \mathbb{R}^D$. Then the data covariance matrix is given as $S = \frac{1}{N}XX^T$



$$Sb_m = \frac{1}{N}XX^Tb_m = \lambda_mb_m \quad (7)$$

$$\frac{1}{N}X^TXX^Tb_m = \lambda_mX^Tb_m \quad (8)$$

$$\frac{1}{N}X^T Xc_m = \lambda_m c_m \quad (9)$$

- Here $c_m = X^Tb_m$.
- The nonzeroeigenvalues of XX^T is same as the nonzero eigenvalues of X^TX .



- Now that we have the eigenvectors of $\frac{1}{N}X^T X$,
- We need to derive the eigenvectors of XX^T , which we still need for PCA

$$\frac{1}{N}XX^T Xc_m = \lambda_m Xc_m \quad (10)$$

- Here, we recover the data covariance matrix again.
- This now also means that we recover Xc_m as an eigenvector of S .

KEY STEPS OF PCA IN PRACTICE



- In the following, we will go through the individual steps of PCA using a running example.
- We are given a two dimensional dataset and we want to use PCA to project it onto a one-dimensional subspace. The key steps are given below
 - ▶ Mean subtraction
 - ▶ Standardization
 - ▶ Eigendecomposition of the covariance matrix
 - ▶ Projection



Mean subtraction

- 1 We start by centering the data by computing the mean of the dataset and subtracting it from every single data point.
- 2 This ensures that the dataset has mean 0.
- 3 Mean subtraction is not strictly necessary but reduces the risk of numerical problems



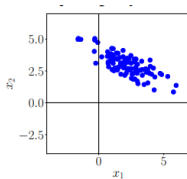
Standardization

- 1 Divide the data points by the standard deviation σ_d of the dataset for every dimension d .
- 2 Now the data is unit free, and it has variance 1 along each axis, which is indicated by the standardization
- 3 This step completes the standardization of the data.

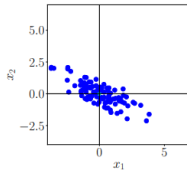
KEY STEPS OF PCA IN PRACTICE



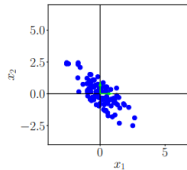
Figure: PCA



(a) Original dataset.



(b) Step 1: Centering by subtracting the mean from each data point.



(c) Step 2: Dividing by the standard deviation to make the data unit free. Data has variance 1 along each axis.

Figure 10.11 Steps of PCA. (a) Original dataset; (b) centering; (c) divide by standard deviation; (d) eigendecomposition; (e) projection; (f) mapping back to original data space.

KEY STEPS OF PCA IN PRACTICE



Eigendecomposition of the covariance matrix

- 1 Compute the data covariance matrix and its eigenvalues and corresponding eigenvectors.
- 2 Since the covariance matrix is symmetric, the spectral theorem states that we can find an orthonormal basis of eigenvectors.
- 3 The eigenvectors are scaled by the magnitude of the corresponding eigenvalue.



Projection

- 1 We can project any data point $x_* \in \mathbb{R}^d$ onto the principal subspace:
- 2 To get this right, we need to standardize x_* using the mean and standard deviation of the training data in the d th dimension

$$x_*^{(d)} = \frac{x_*^{(d)} - \mu_d}{\sigma_d}, \quad d = 1, \dots, D \quad (11)$$

- 3 Here $x_*^{(d)}$ is the d th component of x_* .

KEY STEPS OF PCA IN PRACTICE



- 1 We obtain the projection as

$$\tilde{x} = BB^T x_*$$
(12)

- 2 The coordinates are

$$z_* = B^T x_*$$
(13)

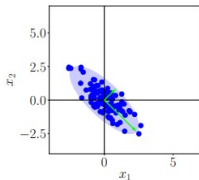
with respect to the basis of the principal subspace.

- 3 Here, B is the matrix that contains the eigenvectors that are associated with the largest eigenvalues of the data covariance matrix as columns.

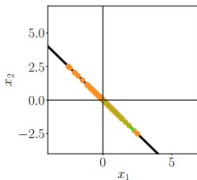
KEY STEPS OF PCA IN PRACTICE



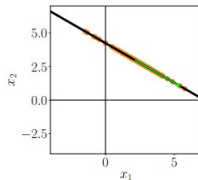
Figure: PCA



(d) Step 3: Compute eigenvalues and eigenvectors (arrows) of the data covariance matrix (ellipse).



(e) Step 4: Project data onto the principal subspace.



(f) Undo the standardization and move projected data back into the original data space from (a).

SUMMARY OF PCA



- 1 We derived PCA from two perspectives: (a) maximizing the variance in the projected space; (b) minimizing the average reconstruction error.
- 2 We took high-dimensional data $x \in \mathbb{R}^D$ and used a matrix B to find a lower-dimensional representation $z \in \mathbb{R}^M$
- 3 The columns of B are the eigenvectors of the data covariance matrix S that are associated with the largest eigenvalues.
- 4 Once we have a low-dimensional representation z , we can get a high-dimensional version of it as Bz .